



US 20250283672A1

(19) **United States**(12) **Patent Application Publication**
TIZIANO et al.(10) **Pub. No.: US 2025/0283672 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **HEAT EXCHANGER COMPRISING A
SEPARATOR ELEMENT**(52) **U.S. Cl.**CPC *F28F 9/0202* (2013.01); *F28F 9/0243*
(2013.01); *F28F 2275/04* (2013.01)(71) Applicant: **DENSO THERMAL SYSTEMS**
S.p.A., Poirino (Torino) (IT)

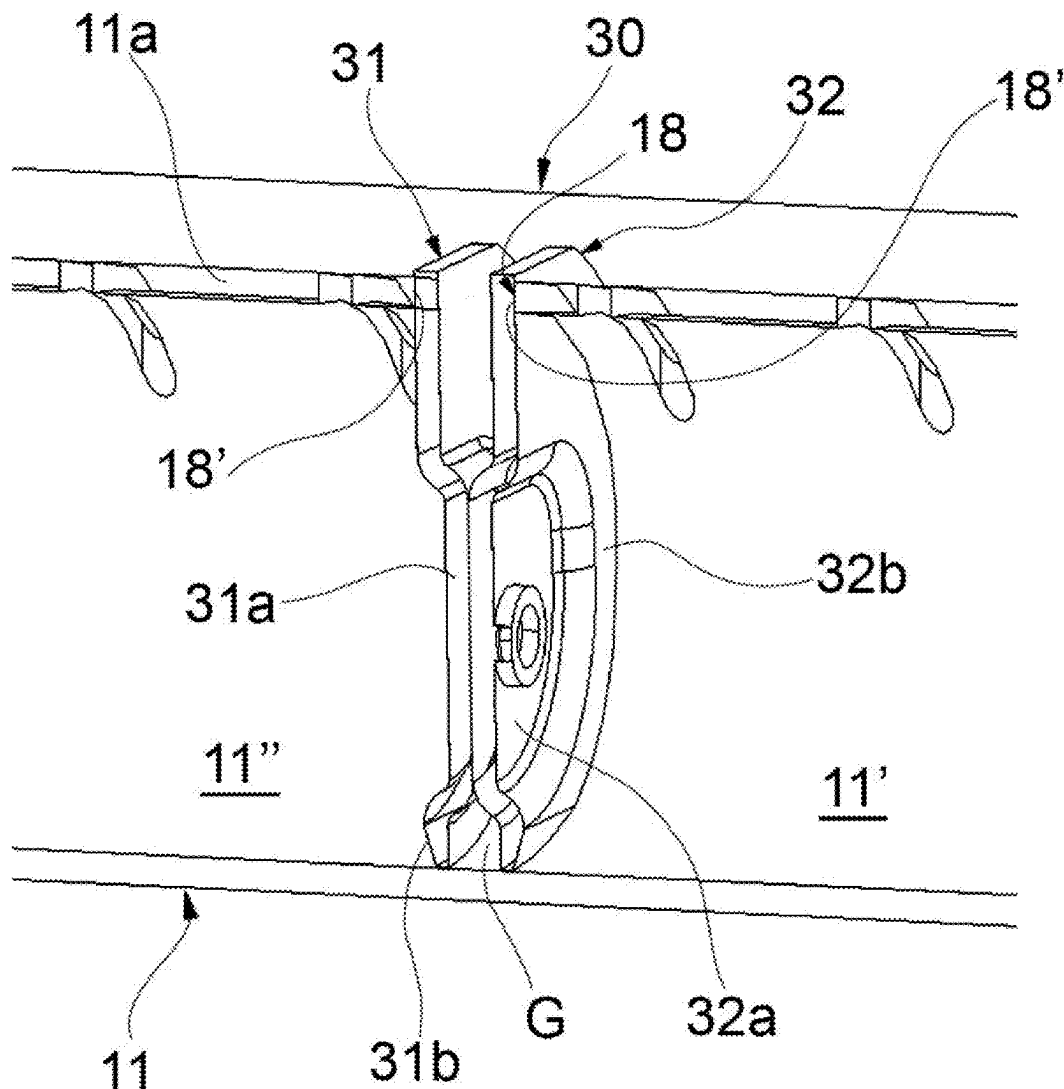
(57)

ABSTRACT(72) Inventors: **Giuseppe TIZIANO**, Poirino (Torino)
(IT); **Fabio SANDRI**, Poirino (Torino)
(IT); **Roberta DREASSI**, Poirino
(Torino) (IT)

A heat exchanger has a first distributor and a second distributor, and a plurality of parallel, coplanar tubes interconnecting the first distributor and the second distributor. One of the first and second distributors has a separator element inserted into a through-hole formed on a side wall of the distributor. The separator element has a pair of plates, each of which has a central portion and a peripheral edge raised with respect to the central portion, the plates being joined to each other at their central portions to define an annular gap around the central portions, interposed between the peripheral edges of the plates. The peripheral edge of each plate has a pair of laterally protruding portions arranged on opposite sides of the plate, and interposed between opposite edges of the through-hole.

(21) Appl. No.: **19/073,246**(22) Filed: **Mar. 7, 2025**(30) **Foreign Application Priority Data**

Mar. 7, 2024 (IT) 102024000005059

Publication Classification(51) **Int. Cl.**
F28F 9/02 (2006.01)

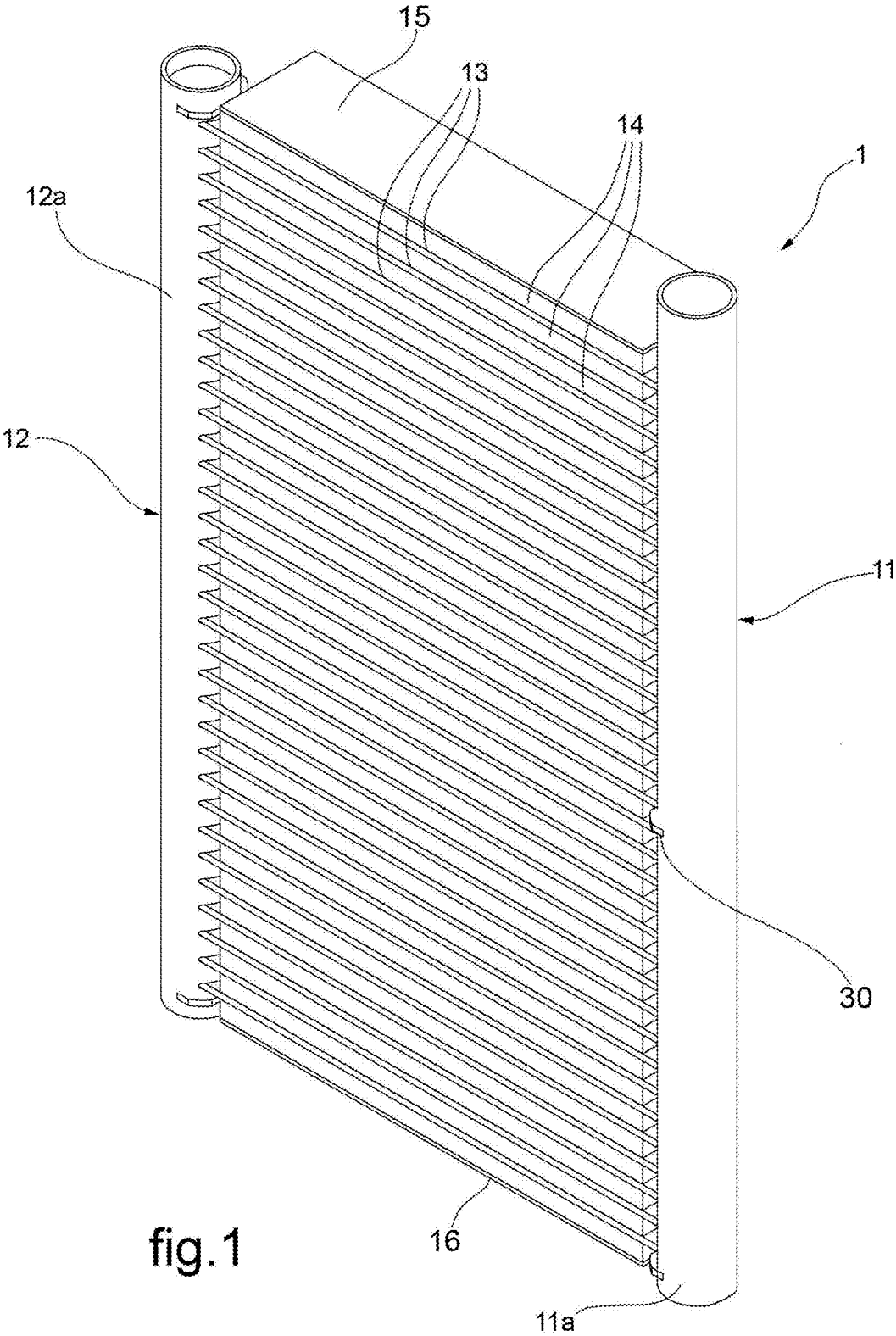


fig.1

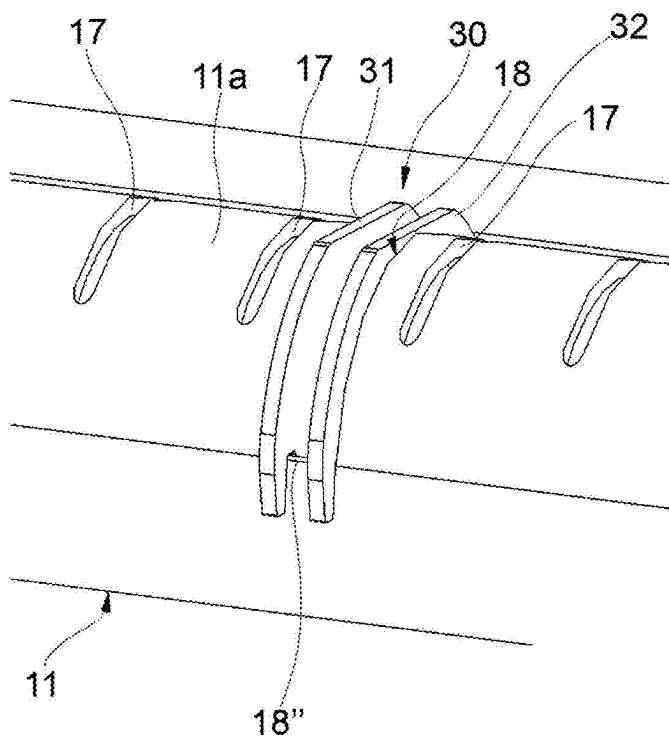


fig.2

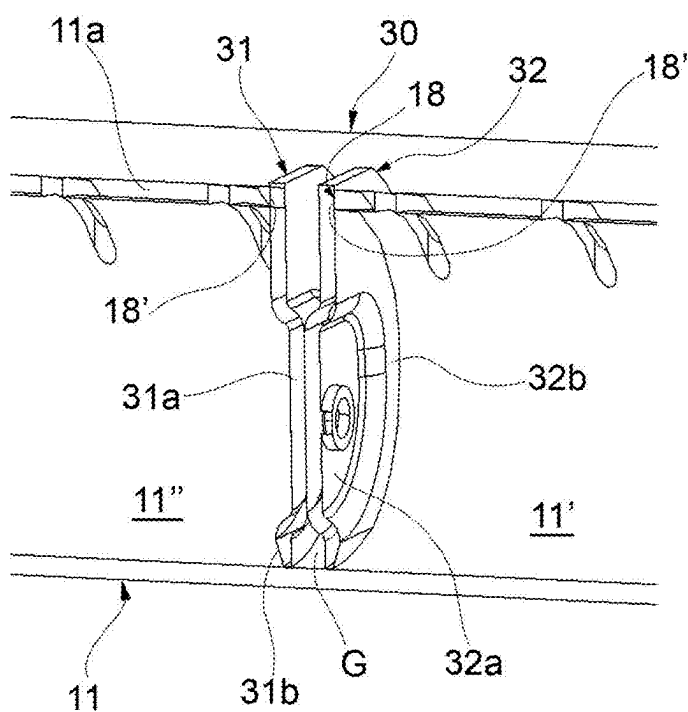
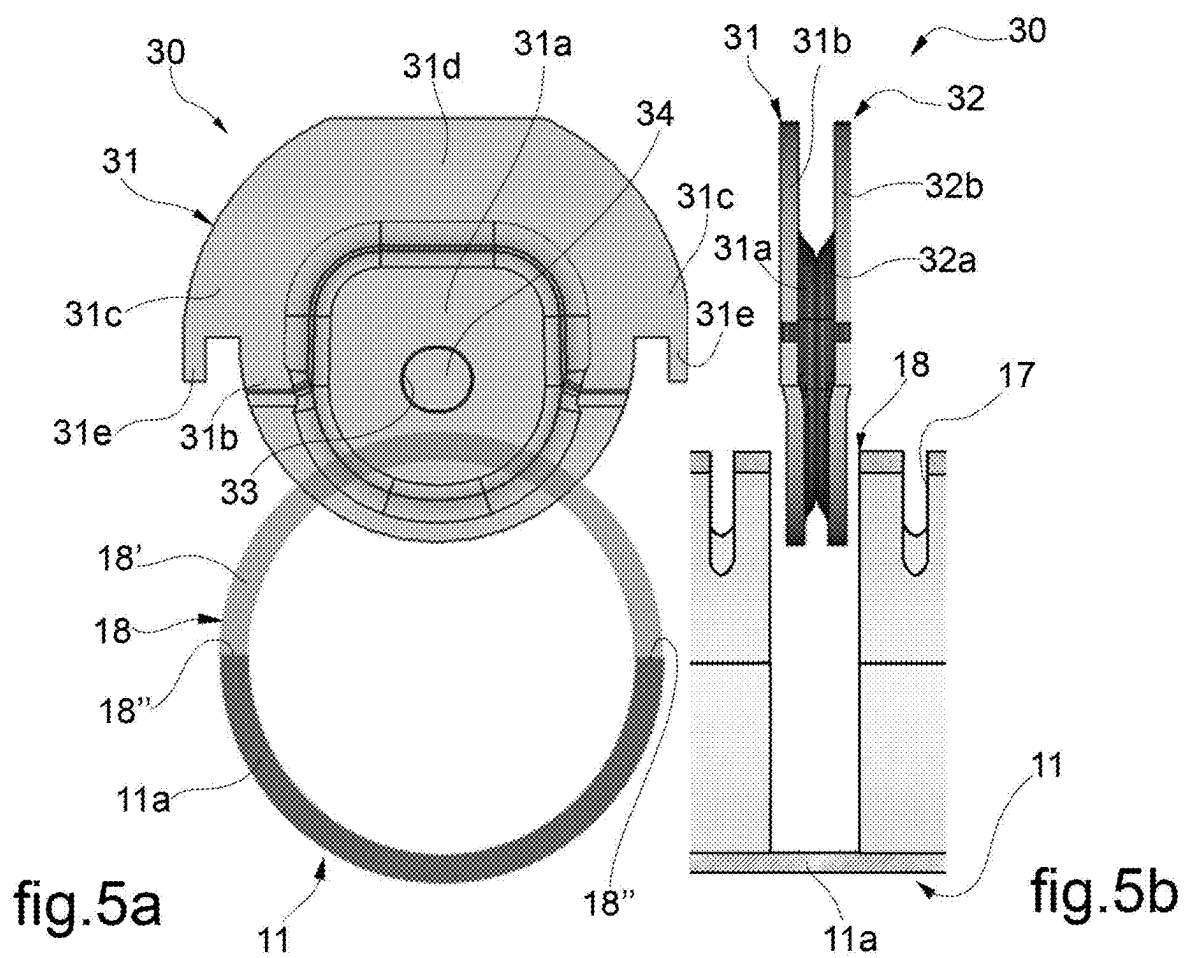
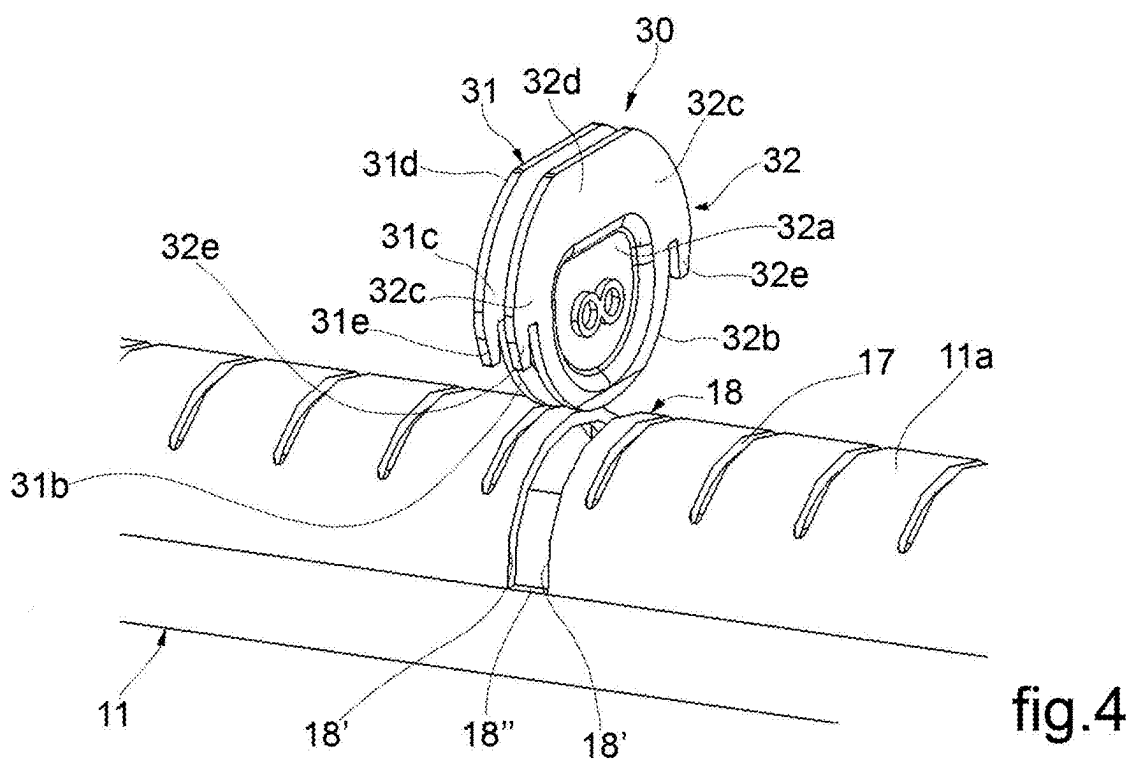
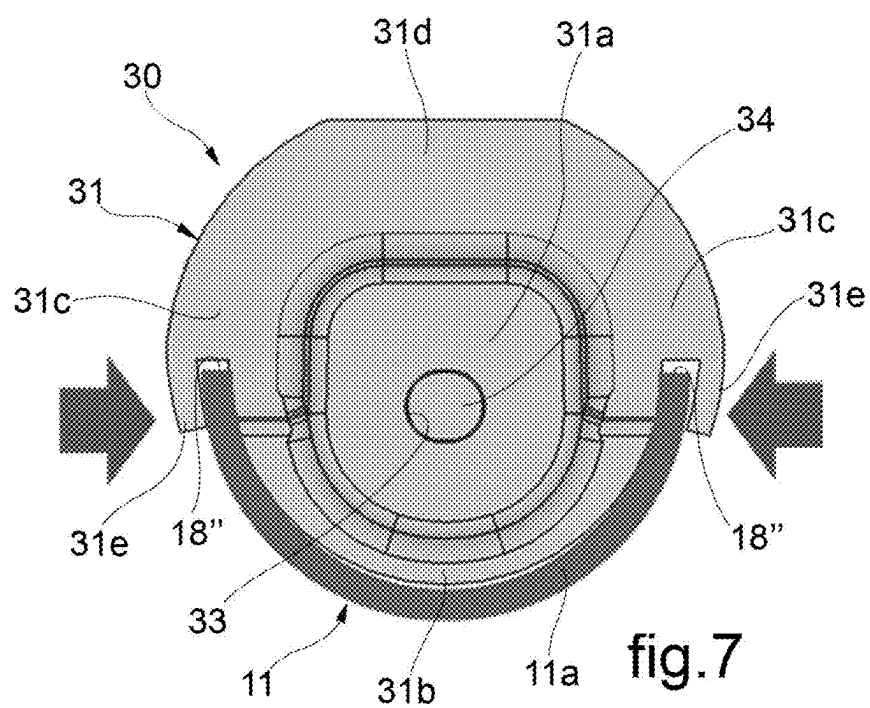
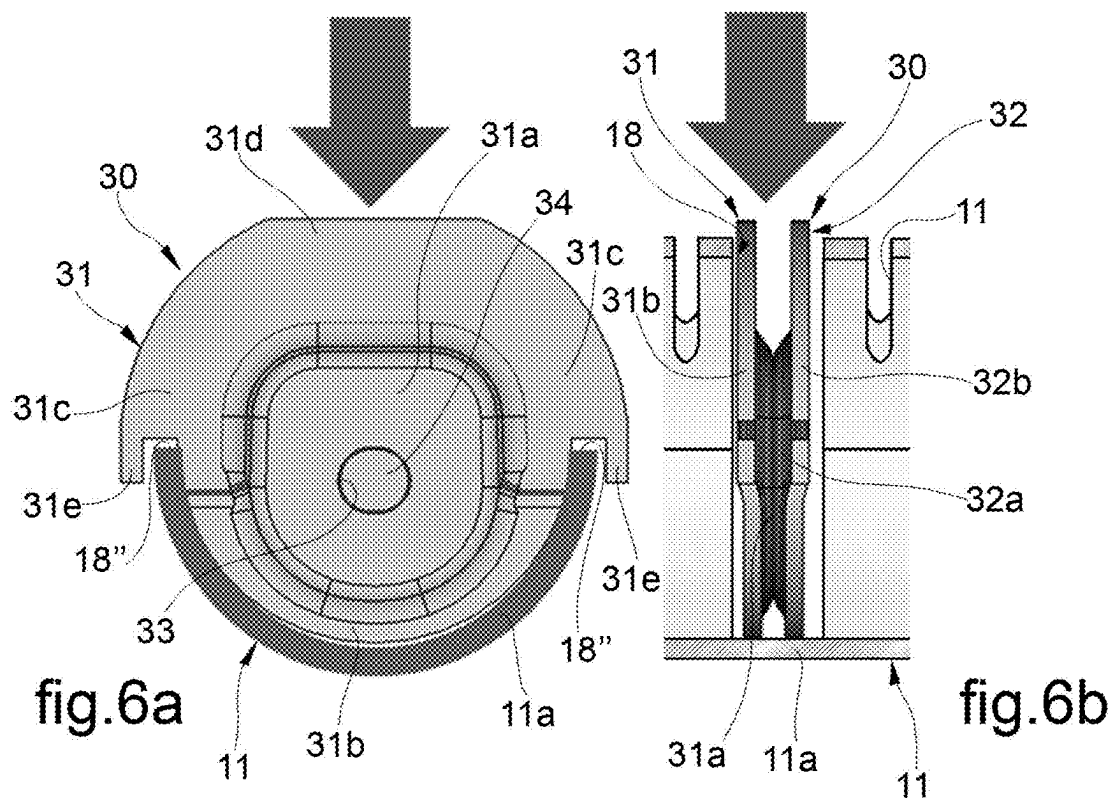
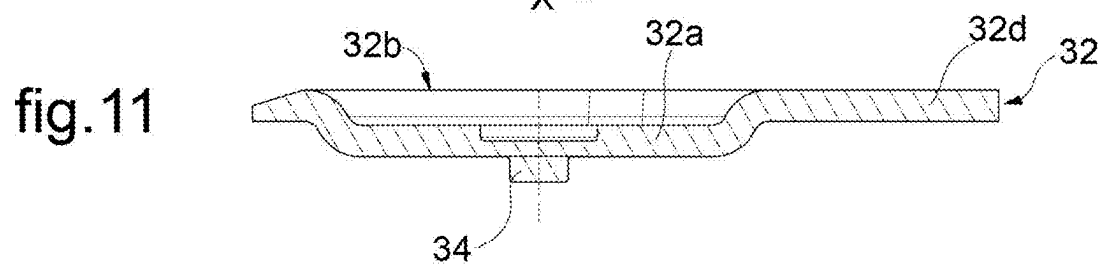
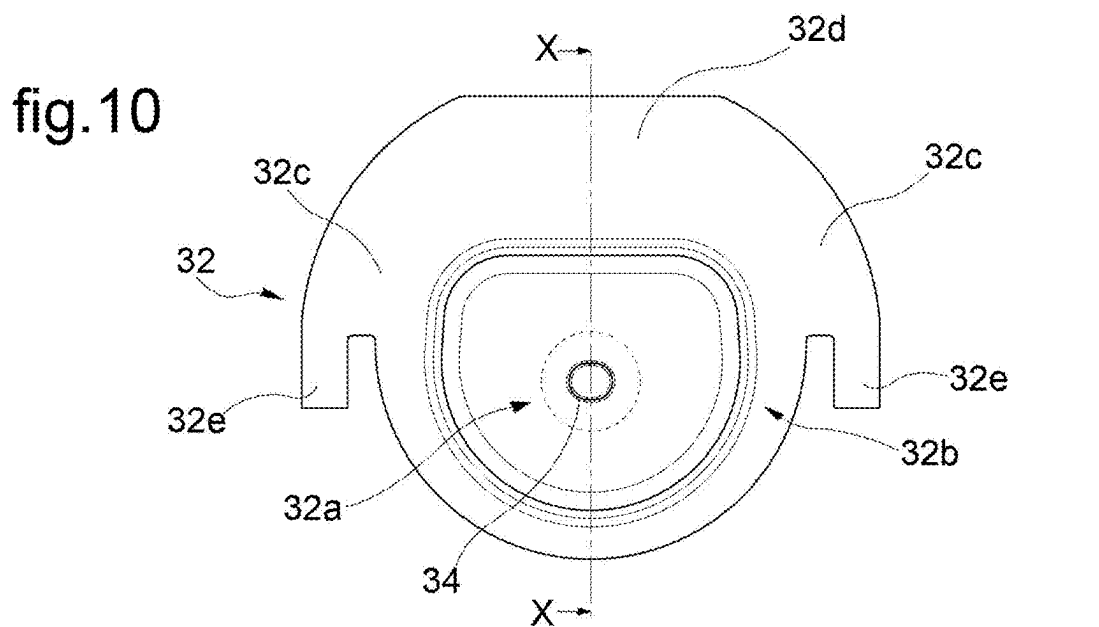
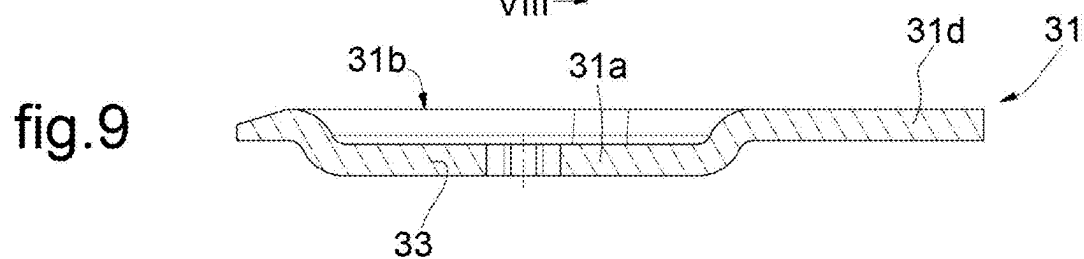
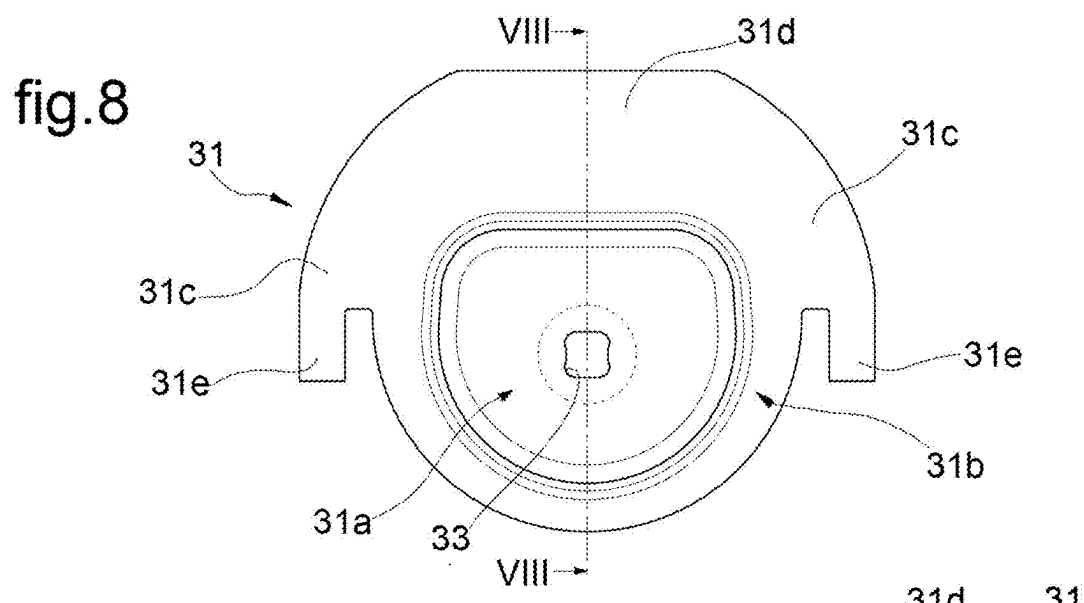
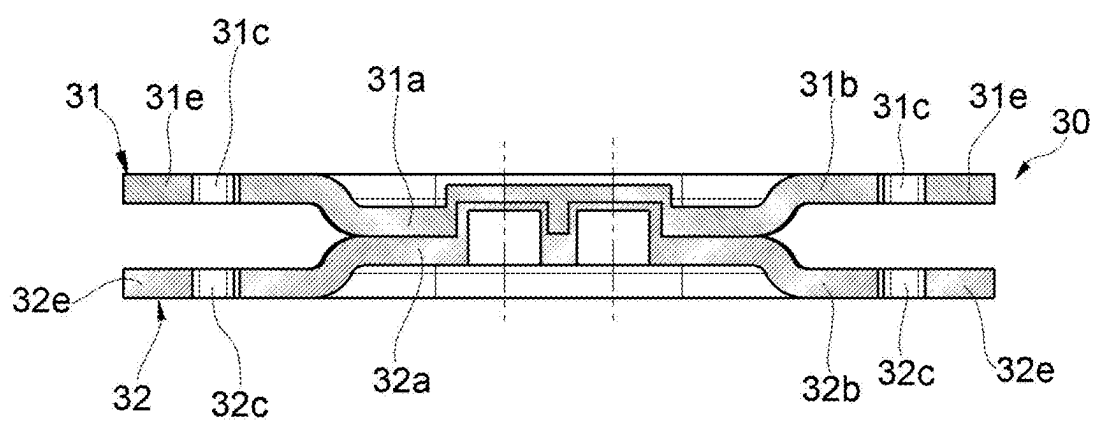
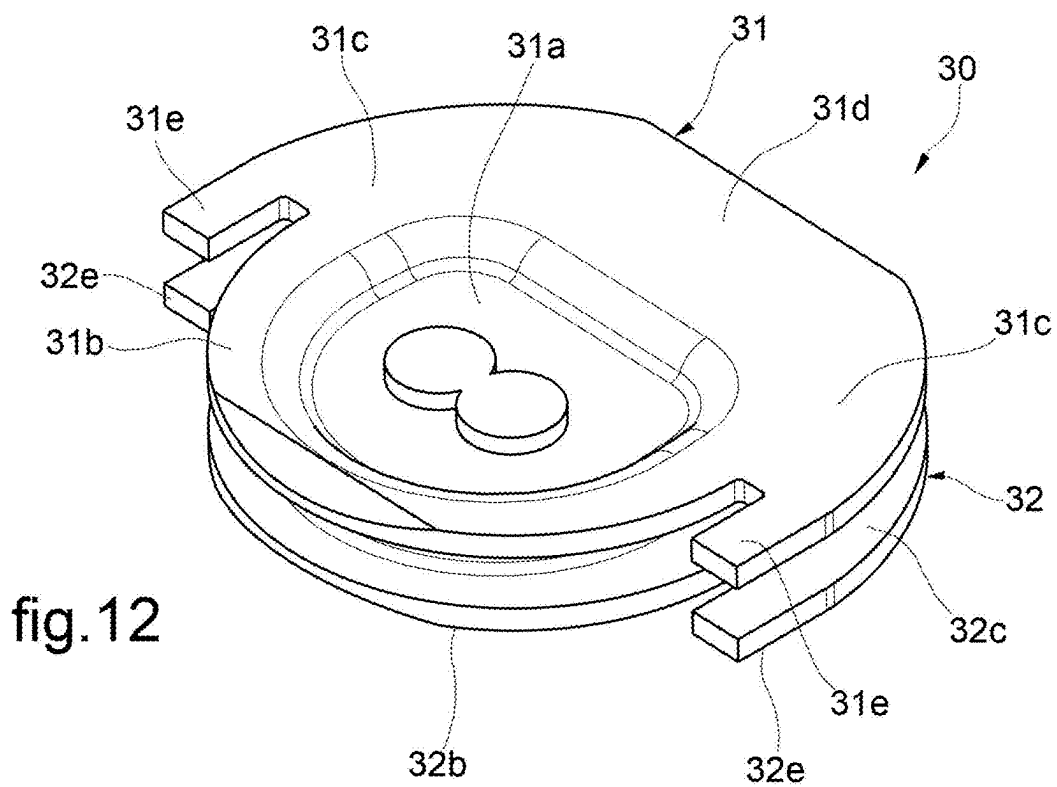


fig.3









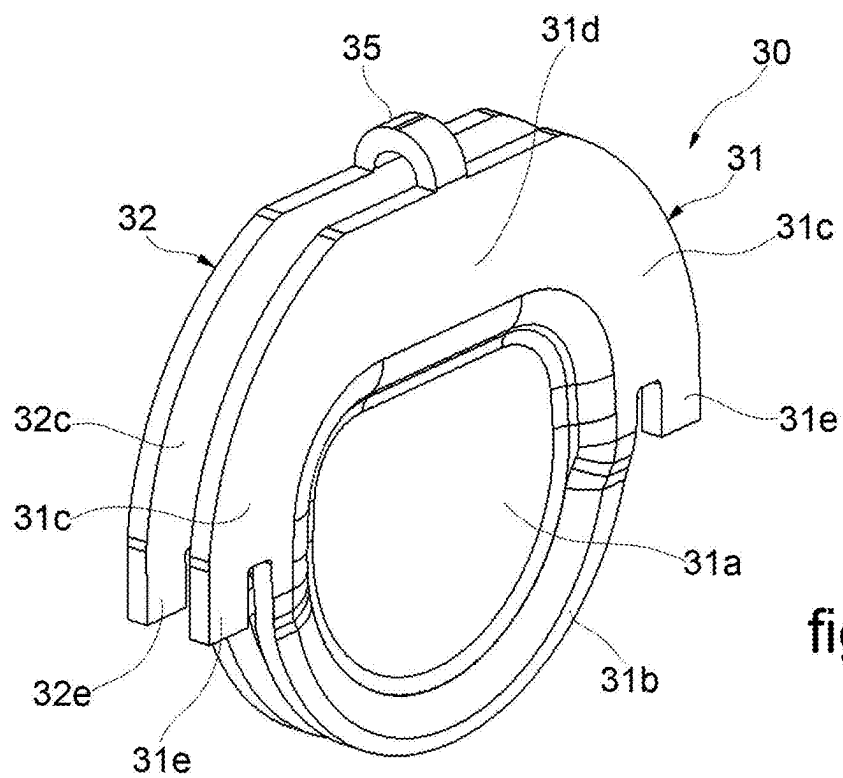


fig.14

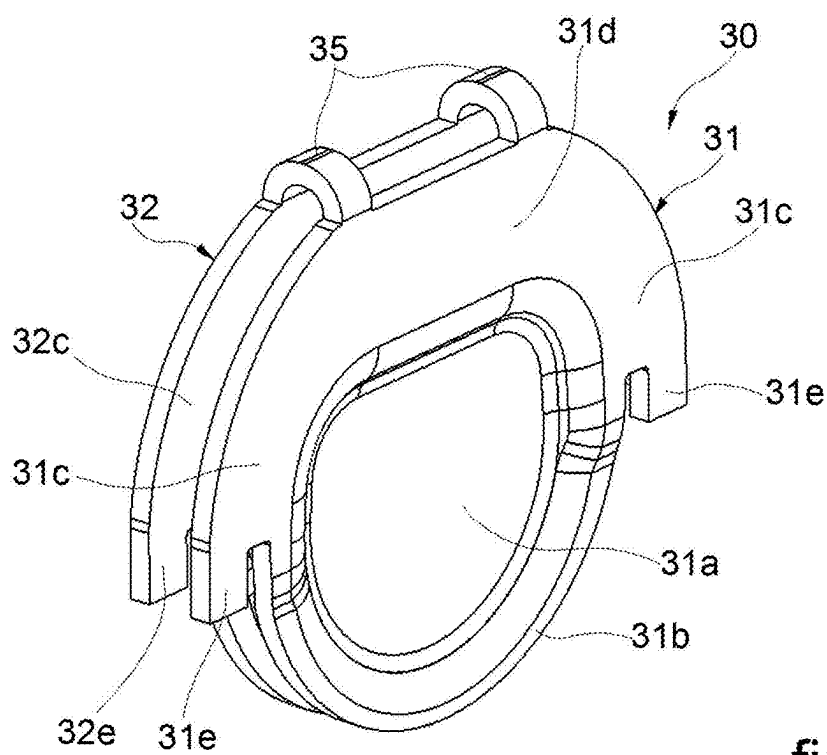


fig.15

HEAT EXCHANGER COMPRISING A SEPARATOR ELEMENT

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Italian Patent Application No. 102024000005059 filed Mar. 7, 2024, the contents of which is hereby incorporated in its entirety by reference.

FIELD OF THE INVENTION

[0002] The present invention relates to a heat exchanger, particularly for a motor vehicle, of the type comprising a first distributor and a second distributor and a plurality of parallel, coplanar tubes interconnecting the first distributor and the second distributor, wherein each of the distributors comprises a tubular body, on a side wall of which a plurality of slits is formed, in each of which an end of a respective tube is inserted, wherein at least one of the distributors, hereinafter “partitioned distributor”, comprises a separator element which partitions the inside of the partitioned distributor into a first inner space and a second inner space, the separator element being inserted into a through-hole formed in the side wall of the partitioned distributor.

BACKGROUND OF THE INVENTION

[0003] Heat exchangers of this type are used, for example, in HVAC systems with a heat pump.

[0004] In such heat exchangers, a separator partition is installed to divide the passages inside the heat exchanger. When there is an internal leakage from the separator (i.e., a seepage between the two internal spaces formed inside the partitioned distributor), this results in a significant performance loss. Detecting such leaks during the production process is highly complex, and the available detection systems often lack the sensitivity to identify small leaks that may negatively affect the performance of the heat exchanger. JP 2016099097 A describes a heat exchanger provided with collectors formed by a plate and a basin assembled together and comprising a separator made up of a pair of plates provided with a protruding structure through a wall of the basin. The plates are joined to each other at the respective central portions so as to define an annular gap around the central portions, interposed between the peripheral edges of the plates. The separator improves leak detection by directing any leakage toward the exterior of the heat exchanger, making it detectable through a simple pressure-tightness test.

SUMMARY OF THE INVENTION

[0005] One object of the present invention is to make a similar solution available for heat exchangers provided with tubular body distributors.

[0006] This object is achieved according to the present invention by a heat exchanger of the type described hereinabove, wherein the separator element comprises a pair of plates, each of which comprises a central portion and a peripheral edge raised with respect to the central portion, the plates being joined to each other at their central portions so as to define an annular gap around the central portions, interposed between the peripheral edges of the plates, and wherein the peripheral edge of each plate comprises a pair

of laterally protruding portions arranged on opposite sides of the plate and interposed between opposite edges of the through-hole.

[0007] Preferred embodiments of the present invention are also described.

BRIEF DESCRIPTION OF THE FIGURES

[0008] Further features and advantages of the present invention will become clearer with the following detailed description of an embodiment of the invention, made with reference to the accompanying drawings, provided purely by way of illustration and not limitation, in which:

[0009] FIG. 1 is a perspective view of a heat exchanger; [0010] FIG. 2 is a perspective view of a detail of a distributor of a heat exchanger according to the present invention;

[0011] FIG. 3 is a sectional perspective view of the detail of FIG. 2;

[0012] FIG. 4 is an exploded view of the detail of FIG. 2;

[0013] FIGS. 5a and 5b are respectively a cross-sectional view and a longitudinal sectional view illustrating the insertion of a separator element into a through-hole of the distributor;

[0014] FIGS. 6a and 6b are respectively a cross-sectional view and a longitudinal sectional view illustrating the separator element completely inserted into the through-hole of the distributor;

[0015] FIG. 7 is a cross-sectional view illustrating a further processing phase;

[0016] FIG. 8 is a plan view of a first plate of the separator element;

[0017] FIG. 9 is a sectional view of the first plate taken along line VIII-VIII of FIG. 8;

[0018] FIG. 10 is a plan view of a second plate of the separator element;

[0019] FIG. 11 is a sectional view of the second plate taken along line X-X of FIG. 10, according to an embodiment;

[0020] FIGS. 12 and 13 are respectively a perspective view and a sectional view of the separator element according to another embodiment; and

[0021] FIGS. 14 and 15 are perspective views of further embodiments.

DETAILED DESCRIPTION

[0022] With reference to FIG. 1, a heat exchanger made of metallic material, particularly aluminum, is shown overall denoted by 1. In the illustrated example, the heat exchanger 1 is a heat exchanger for heat transfer fluid-air, particularly an internal condenser of an HVAC system with a heat pump for an electric vehicle.

[0023] The heat exchanger 1 comprises at least a first distributor 11 and at least a second distributor 12 and a plurality of parallel, coplanar tubes 13 interconnecting the first distributor 11 and the second distributor 12. Between adjacent tubes 13, fins 14 are interposed, not shown in detail. Two lateral plates 15 and 16 are fixed to opposite sides of the mass formed by the tubes 13.

[0024] The above-described elements are joined together in a per se known manner, for example, by braze-welding.

[0025] Each of the distributors 11, 12 comprises a tubular body, for example, with a circular cross-section, on a side wall 11a, 12a of which a plurality of slits 17 is formed, in

each of which an end of a respective tube **13** is inserted. As shown in FIGS. **1** to **4**, inside at least one of the distributors **11**, **12**, a

[0026] separator element **30** is arranged, partitioning the inside of the partitioned distributor **11** into a first inner space **11'** and a second inner space **11''**. The separator element **30** is inserted into a linear through-hole **18** formed on the side wall **11a** of the partitioned distributor **11** and parallel to the slits **17** for the tubes **13**. The separator element **30** comprises an edge abutting against an internal surface of the partitioned distributor **11** and opposite faces contiguous with respective edges **18'** of the through-hole **18**. The separator element **30** is joined to the partitioned distributor **11** by material coupling, particularly by braze-welding.

[0027] With reference to FIGS. **8** to **11**, the separator element **30** comprises a pair of plates **31**, **32**, each of which comprises a central portion **31a**, **32a** and a peripheral edge **31b**, **32b** raised with respect to the central portion **31a**, **32a**. As can be seen in FIGS. **3-4**, the plates **31**, **32** are joined to each other at their respective central portions **31a**, **32a** so as to define an annular gap **G** around the central portions **31a**, **32a**. The annular gap **G** is interposed between the peripheral edges **31b**, **32b** of the plates **31**, **32** and, therefore, communicates with the external environment through the portion of the through-hole **18** located between the peripheral edges **31b**, **32b** of the plates **31**, **32**. Any fluid seepage between the edges of the separator element **30** and the internal surface of the partitioned distributor **11** is thus detectable through the through-hole **18** via a tightness test.

[0028] The peripheral edge **31b**, **32b** of each plate **31**, **32** comprises a pair of laterally protruding portions **31c**, **32c** arranged on opposite sides of the plate **31**, **32** and interposed between opposite edges **18'** of the through-hole **18**. The laterally protruding portions **31c**, **32c** are connected to each other by a central protruding portion **31d**, **32d** of the peripheral edge **31b**, **32b**. The central protruding portions **31d**, **32d** are also interposed between the opposite edges **18'** of the through-hole **18**. As can be seen in FIGS. **2** and **3**, the laterally protruding portions **31c**, **32c** and central protruding portions **31d**, **32d** of the plates **31**, **32** extend outward beyond the through-hole **18**, thus protruding with respect to the outer surface of the partitioned distributor **11**. The laterally protruding portions **31c**, **32c** of the plates **31**, **32** overlap opposite ends **18''** of the through-hole **18** and are preferably provided with respective crimping appendages **31e**, **32e**.

[0029] Preferably, the plates **31-32** are directly fixed to each other so as to achieve a provisional assembly of the plates **31**, **32** before their insertion into the through-hole **18**. In the example illustrated in FIGS. **8-11**, one of the plates **31**, **32** is provided with a hole **33** formed through its respective central portion **31a**, and the other of the plates **31**, **32** is provided with a button-like protrusion **34** formed on its respective central portion **32a**. The hole **33** and the button-like protrusion **34** engage with each other to achieve the provisional assembly of the plates **31**, **32** before their insertion into the through-hole **18**.

[0030] As shown in FIGS. **5a-5b** and **6a-6b**, the separator element **30**, formed by assembled plates **31**, **32**, is inserted from the outside of the distributor **11** through the through-hole **18**, until the edges of the plates **31**, **32** abut against the internal surface of the distributor **11** and the laterally protruding portions **31c**, **32c** overlap ends **18''** of the through-hole **18**. Subsequently, the crimping appendages **31e**, **32e** are

deformed against the external surface of the distributor **11**, thereby achieving a provisional attachment of the separator element **30** to the distributor **11**. Finally, braze-welding is used to achieve the definitive joining of the separator element **30** to the distributor **11**.

[0031] FIGS. **12** and **13** show a further embodiment of the present invention. Elements corresponding to those of the previous embodiment have been assigned the same reference numbers; these elements will not be further described. The embodiment of FIGS. **12** and **13** differs from the previous one in the type of engagement between the plates **31**, **32** of the separator element **30**. In this embodiment, the engagement between the plates **31**, **32** is achieved by cold deformation, which creates undercuts in the central portions **31a**, **32a**.

[0032] FIGS. **14** and **15** show further embodiments of the present invention. Elements corresponding to those of the previous embodiments have been assigned the same reference numbers; these elements will not be further described. The embodiments of

[0033] FIGS. **14** and **15** differ from the previous ones in that the plates **31**, **32** of the separator element **30** are connected to each other by one or more connection portions **35** made in one piece with the plates **31**, **32**. The connection portions **35** extend between the central protruding portions **31d**, **32d** of the peripheral edge **31b**, **32b** of the plates **31**, **32**. In this embodiment, the connection portions **35** are bent before inserting the separator element **30** into the through-hole **18**, so as to arrange the plates **31**, **32** with the central portions **31a**, **32a** against each other.

What is claimed is:

1. A heat exchanger comprising a first distributor and a second distributor, and a plurality of parallel, coplanar tubes interconnecting the first distributor and the second distributor,

wherein each of the first and second distributors comprises a tubular body, on a side wall of which a plurality of slits is formed,

wherein in each slit of the plurality of slits an end of a respective tube of the plurality of parallel, coplanar tubes is inserted,

wherein at least one of the first and second distributors, hereinafter "partitioned distributor", comprises a separator element that partitions an inside of the partitioned distributor into a first inner space and a second inner space, the separator element being inserted into a through-hole formed in the side wall of the partitioned distributor,

wherein the separator element comprises a pair of plates, each of which comprises a central portion and a peripheral edge raised with respect to the central portion, the plates being joined to each other at their central portions so as to define an annular gap around the central portions, interposed between the peripheral edges of the plates, and

wherein the peripheral edge of each plate comprises a pair of laterally protruding portions arranged on opposite sides of the plate, and interposed between opposite edges of the through-hole.

2. The heat exchanger of claim 1, wherein, in each plate, the laterally protruding portions are connected to each other by a central protruding portion of the peripheral edge, and wherein the central protruding portions of the plates are interposed between the opposite edges of the through-hole.

3. The heat exchanger of claim 1, wherein the laterally protruding portions of the plates overlap the opposite edges of the through-hole.

4. The heat exchanger of claim 3, wherein the laterally protruding portions have respective crimping appendages provided for attachment to the tubular body of the partitioned distributor.

5. The heat exchanger of claim 1, wherein the plates are joined to the tubular body of the partitioned distributor by braze-welding.

6. The heat exchanger of claim 1, wherein the plates are directly fixed to each other.

7. The heat exchanger of claim 1, wherein the plates are connected to each other by at least one connection portion made in one piece with the plates, the at least one connection portion being bent to arrange the plates with the central portions against each other.

* * * * *